

LambdaScript : A Functional Programming Language

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Abstract—We present LambdaScript, a statically-typed functional programming language featuring Hindley-Milner type inference, polymorphic algebraic data types, and comprehensive pattern matching. LambdaScript combines the expressiveness of languages like OCaml and Haskell with a clean syntax and rigorous formal semantics. The language supports advanced features including mutually recursive types, exhaustiveness checking for pattern matching, and first-class functions with closures. We describe the formal type system, operational semantics, and implementation of a complete interpreter with over 700 unit tests demonstrating correctness.

Index Terms—functional programming, type inference, algebraic data types, pattern matching, programming languages